ThreatMatrix AI — Day 18 Plan: Frontend Overhaul & VPS Integration

> **Date:** March 31, 2026 (Day 18) **Phase:** Week 6 — Feature Depth Completion + UI Polish **Owner:** Lead Architect (acting as code reviewer + integrator) **v0.5.0 Status:** 100% backend API (46/46 endpoints), Frontend partially wired

---

1. Executive Summary

The backend is **fully operational** on VPS with 46/46 API endpoints verified. The frontend has **10 pages** but only 5 are connected to live APIs (War Room, Alerts, AI Analyst, ML Ops, Intel). The remaining 5 (Forensics, Reports, Hunt, Admin, Network timeline) use mock/static data. Additionally, 6 pages lack AuthGuard protection, and CSS polish is incomplete on several pages.

Goal: Transform the frontend from "partially wired demo" to "fully integrated production UI" by:

1. Connecting to VPS backend from local dev environment
2. Wiring all 5 mock pages to live APIs
3. Adding AuthGuard to all unprotected pages
4. Polishing CSS consistency across all pages
5. Fixing all known bugs from the audit report

---

2. Current State Assessment

2.1 Pages Using Live Data (5/10)

Page	Status	Data Source
`/war-room`	☑ Live	useWebSocket, useFlows, useAlerts, useMLModels, POST /llm/chat
`/alerts`	☑ Live	useAlerts hook with severity/status filters
`/ai-analyst`	☑ Live	useLLM hook, GET /alerts/{id}, GET /llm/budget
`/ml-ops`	☑ Live	useMLModels, mlService.getComparison(), POST /ml/retrain
`/intel`	⚠ Hybrid	useIntel + fallback MOCK_IOCS, live feed status + lookup

2.2 Pages Using Mock Data (5/10)

Page	AuthGuard	Data Source	Backend Endpoint Available
`/forensics`	✗ Missing	MOCK_UPLOADS (3 entries)	POST /capture/upload-pcap ☑
`/reports`	✗ Missing	MOCK_REPORTS (4 entries)	POST /reports/generate, GET /reports/, GET /reports/{id}/download ☑

Page	AuthGuard	Data Source	Backend Endpoint Available
`/hunt`	❌ Missing	MOCK_FLOWS (5 entries)	GET /flows, POST /flows/search ☑
`/admin`	❌ Missing	Hardcoded ADMIN_CARDS	GET /admin/audit-log, GET /llm/budget ☑
`/network`	❌ Missing	useFlows (live) + MOCK_FLOWS fallback	Timeline data empty (GET /flows/stats?interval=1m) ☑

2.3 CSS Issues Found

Issue	Severity	Files Affected
`page-container` class used but not defined in globals.css	●	forensics, reports, hunt, admin
`@keyframes spin` missing (ML Ops retrain button broken)	●	ml-ops/page.tsx
Inline styles dominate all pages (no reusable CSS classes)	○	All pages
No focus-visible ring styles for inputs	○	hunt, intel, network
Inconsistent page header patterns	○	All pages
No hover-lift effect on Admin cards	●	admin/page.tsx

2.4 Bug Inventory from Audit

Bug	File	Status
APP_VERSION is 'v0.1.0' — should be 'v0.5.0'	constants.ts:79	Open
AuthGuardWrapper is a no-op (`return <>{children}</>`)	AuthGuardWrapper.tsx	Open
ML Ops hyperparameter values hardcoded	ml-ops/page.tsx	Open
TrafficTimeline receives `data={[]}`	war-room/page.tsx, network/page.tsx	Open
GeoDistribution is fully static	war-room, network pages	Open
Forensics upload has no onClick handler	forensics/page.tsx	Open
Reports buttons are decorative (no handlers)	reports/page.tsx	Open
Hunt export/analyze buttons have no handlers	hunt/page.tsx	Open
Admin cards have cursor:pointer but no onClick	admin/page.tsx	Open

3. Environment Setup Strategy

3.1 Local Frontend → VPS Backend Connection

The frontend runs locally (`pnpm dev` on port 3000) and connects to the VPS backend at `http://187.124.45.161:8000`. The backend has CORS configured for `localhost:3000` and `localhost:3001`.

Step 1: Create `.env.local`

```
NEXT_PUBLIC_API_URL=http://187.124.45.161:8000
NEXT_PUBLIC_WS_URL=ws://187.124.45.161:8000
```

Step 2: DEV_MODE Bypass The VPS backend has `DEV_MODE=true`, so we can set a token in browser console:

```
localStorage.setItem('tm_access_token', 'dev-mode-bypass');
window.location.reload();
```

Step 3: Verify Connectivity

```
# Test API reachability
curl http://187.124.45.161:8000/api/v1/system/health

# Start frontend
cd frontend
pnpm dev --port 3000
```

---

4. Task Breakdown (Ordered by Priority)

Task 1: Fix AuthGuard + Missing CSS (🔴 Critical — 30 min)

4.1.1 — Restore AuthGuardWrapper File: `frontend/components/auth/AuthGuardWrapper.tsx`

Current (no-op):

```
export function AuthGuardWrapper({ children }: { children: React.ReactNode }) {
  return <>{children}</>;
}
```

Fix: Restore the full implementation from INTEGRATION_WALKTHROUGH_FINAL.md (the original was created there but later replaced with a no-op). The implementation should:

- Check `localStorage.getItem('tm_access_token')`
- Redirect to `/login` if missing and not on login page
- Show nothing while checking
- Allow login page to render without auth

4.1.2 — Add Missing `page-container` CSS Class File: `frontend/app/globals.css`

Add after `.main-content`:

```
.page-container {
  display: flex;
  flex-direction: column;
  gap: var(--space-4);
}
```

4.1.3 — Add Missing `@keyframes spin` File: `frontend/app/globals.css`

Add with other keyframes:

```
@keyframes spin {
  from { transform: rotate(0deg); }
  to   { transform: rotate(360deg); }
}
```

4.1.4 — Add Focus Ring Styles File: `frontend/app/globals.css`

```
input:focus-visible,
textarea:focus-visible,
select:focus-visible {
  outline: 2px solid var(--cyan);
  outline-offset: 2px;
}
```

Verification:

- ☐ AuthGuard redirects unauthenticated users to /login
- ☐ Login page renders without auth
- ☐ Forensics, Reports, Hunt, Admin pages render without broken layout
- ☐ ML Ops retrain button shows spinner animation
- ☐ Input fields show cyan focus ring

Task 2: Wire Forensics Page to Live API (🔴 Critical — 45 min)

4.2.1 — Connect PCAP Upload File: `frontend/app/forensics/page.tsx`

Changes:

- Import `useCapture` hook and `api` from lib
- Add `<input type="file" ref={fileInputRef} accept=".pcap,.pcapng" hidden />`
- Wire drop zone `onClick` to trigger file input

- Wire `onChange` to call `api.upload('/api/v1/capture/upload-pcap', formData)`
- Show upload progress state
- On success, refetch upload list

4.2.2 — Load Real Upload History

- Fetch `GET /api/v1/capture/status` for current session stats
- Fetch `GET /api/v1/flows?source=pcap` for PCAP-processed flows
- Display real packet/flow/anomaly counts

4.2.3 — Add Drag-and-Drop Feedback

- Add `onDragOver` / `onDragLeave` state for visual feedback
- Change border color to cyan when dragging over

Verification:

- ☐ Click drop zone opens file picker
- ☐ Can upload a .pcap file to VPS
- ☐ Upload progress shown
- ☐ Upload history shows real data from VPS

Task 3: Wire Reports Page to Live API (🔴 Critical — 45 min)

4.3.1 — Connect Generate Button File: `frontend/app/reports/page.tsx`

Changes:

- Import `api` from lib
- Wire "Generate Report" button to `POST /api/v1/reports/generate`
- Show type selector (daily/incident/executive/ml_performance)
- Show loading state during generation
- On success, refetch report list

4.3.2 — Load Real Reports

- Fetch `GET /api/v1/reports/` on mount (and after generation)
- Display real report data (title, type, generated_at, file_size)

4.3.3 — Connect Download Buttons

- Wire "PDF" download buttons to `GET /api/v1/reports/{id}/download`
- Trigger browser download with correct filename

Verification:

- ☐ Generate Report button creates a PDF on VPS
- ☐ Report list shows real generated reports
- ☐ Download button downloads actual PDF file

Task 4: Wire Hunt Page to Live API (🟡 High — 30 min)

4.4.1 — Connect to Flow Search File: `frontend/app/hunt/page.tsx`

Changes:

- Import `useFlows` hook (already has `searchFlows` method)
- Replace `MOCK_FLOWS` with `useFlows` data
- Wire search input to `searchFlows()` with IP/label query
- Wire protocol filter to `searchFlows()` with protocol param
- Show loading state during search

4.4.2 — Wire Export CSV Button

- Collect filtered results into CSV format
- Trigger browser download of `.csv` file

4.4.3 — Wire Analyze with AI Button

- Navigate to `/ai-analyst` with selected flow context as query param

Verification:

- ☐ Search returns real flows from VPS
- ☐ Protocol filter narrows results
- ☐ Export CSV downloads a valid file
- ☐ Analyze with AI navigates to AI Analyst page

Task 5: Wire Admin Page to Live Data (🟡 High — 45 min)

5.5.1 — Load Real Stats File: `frontend/app/admin/page.tsx`

Changes:

- Fetch `GET /api/v1/admin/audit-log` for audit entry count
- Fetch `GET /api/v1/llm/budget` for real LLM spend
- Fetch `GET /api/v1/system/health` for system status
- Display real counts instead of hardcoded strings

5.5.2 — Wire Card Navigation

- Add `onClick` handlers using `useRouter().push(card.href)`
- Cards that have no sub-page yet: show "Coming Soon" toast

Verification:

- ☐ Audit Log card shows real entry count from VPS
- ☐ LLM Budget card shows real spend from VPS
- ☐ Cards navigate to sub-pages (or show coming soon)

Task 6: Fix TrafficTimeline Empty Data (🟡 High — 20 min)

6.6.1 — Connect Timeline to Flow Stats Files: `frontend/app/war-room/page.tsx`, `frontend/app/network/page.tsx`

The `useFlows` hook already fetches `flowService.getStats('1h')` which returns `FlowStatsResponse`. The `TrafficTimeline` component expects `FlowTimeline[]`. Need to:

- Check if `stats` from `useFlows` contains timeline data
- If yes, transform and pass to `TrafficTimeline`
- If the API returns aggregated stats (not timeline), use `flows` array to build a client-side timeline by grouping flows by timestamp into 1-minute buckets

Verification:

- ☐ War Room TrafficTimeline shows non-empty chart
- ☐ Network page TrafficTimeline shows non-empty chart

Task 7: Fix ML Ops Hardcoded Data (🟠 High — 30 min)

7.7.1 — Remove Hardcoded Hyperparameter Tuning File: `frontend/app/ml-ops/page.tsx`

Changes:

- Fetch `GET /api/v1/ml/training-history` for real training data
- Fetch `GET /api/v1/ml/models` for model-specific hyperparams
- Display real values instead of hardcoded "Current F1: 78.75% → Tuned F1: 83.03%"

7.7.2 — Remove Hardcoded Training Config

- Replace hardcoded "114.1s" / "125,973 train samples" with data from `ml_models` table

Verification:

- ☐ Hyperparameter section shows real data from VPS
- ☐ Training config shows real model metadata

Task 8: CSS Polish Pass (🟢 Low — 60 min)

8.8.1 — Page Header Standardization Create reusable CSS class in globals.css:

```
.page-header {
  display: flex;
  justify-content: space-between;
  align-items: center;
  margin-bottom: var(--space-6);
}

.page-title {
  font-family: var(--font-heading);
  font-size: var(--text-2xl);
  font-weight: 700;
```

```
  color: var(--text-primary);
}

.page-subtitle {
  color: var(--text-muted);
  font-size: var(--text-sm);
  margin-top: 4px;
}
```

8.8.2 — Admin Card Hover Effect Add hover lift to GlassPanel or add specific CSS:

```
.admin-card:hover {
  transform: translateY(-2px);
  box-shadow: 0 8px 24px rgba(0, 0, 0, 0.3);
}
```

8.8.3 — Hunt Search Focus Ring Ensure search input uses focus-visible styles (from Task 1).

8.8.4 — Forensics Drop Zone Hover State Add CSS for drag-hover state:

```
.drop-zone--active {
  border-color: var(--cyan) !important;
  background: var(--cyan-muted) !important;
}
```

Verification:

- ☐ All pages use consistent header pattern
- ☐ Admin cards have hover lift effect
- ☐ Forensics drop zone shows cyan border on drag
- ☐ All inputs show focus ring

Task 9: Constants & Version Cleanup (🟢 Low — 5 min)

9.9.1 — Update APP_VERSION File: `frontend/lib/constants.ts`

```
export const APP_VERSION = 'v0.5.0';  // was 'v0.1.0'
```

Verification:

- ☐ StatusBar shows "v0.5.0"

Task 10: Final TypeScript Verification (🟢 Low — 10 min)

10.10.1 — Run Type Check

```
cd frontend
pnpm tsc --noEmit
```

10.10.2 — Fix Any `any` Types

- `FilterGroup` in alerts page uses `any` — add proper type
- `useAlerts.ts` filters uses `any` — add proper type

Verification:

- ☐ `pnpm tsc --noEmit` passes with 0 errors

---

5. Execution Order

Step	Task	Duration	Cumulative
1	Environment setup (.env.local, dev-mode token)	10 min	10 min
2	Task 1: AuthGuard + Missing CSS	30 min	40 min
3	Task 2: Wire Forensics	45 min	85 min
4	Task 3: Wire Reports	45 min	130 min
5	Task 4: Wire Hunt	30 min	160 min
6	Task 5: Wire Admin	45 min	205 min
7	Task 6: Fix TrafficTimeline	20 min	225 min
8	Task 7: Fix ML Ops hardcoded	30 min	255 min
9	Task 8: CSS polish pass	60 min	315 min
10	Task 9: Version cleanup	5 min	320 min
11	Task 10: TypeScript verification	10 min	330 min
Total		**~5.5 hours**	

---

6. Testing Checklist (Per Audit Report §6.2)

#	Page	Test	Expected Result
1	`/login`	Enter credentials, click Sign In	Auth error or redirect to war-room
2	`/war-room`	Check metric cards	Packets/sec + Active flows update
3	`/war-room`	Watch for anomaly toasts	Toast popup on ML anomaly
4	`/war-room`	Check Threat Map	Deck.gl renders with map

#	Page	Test	Expected Result
5	`/war-room`	Check AI Briefing	LLM-generated briefing appears
6	`/war-room`	Check TrafficTimeline	Chart shows non-empty data
7	`/alerts`	Check alert table	Live alerts with severity badges
8	`/alerts`	Click alert row	AlertDetailDrawer opens
9	`/alerts`	Click "Analyze with AI"	Navigates to /ai-analyst?alert_id=...
10	`/ai-analyst`	Type message, Enter	LLM responds with streaming text
11	`/ai-analyst`	Check budget widget	Shows request count + tokens
12	`/ml-ops`	Check model cards	3 models with real metrics
13	`/ml-ops`	Click "RETRAIN MODELS"	Spinner + status polling
14	`/ml-ops`	Check hyperparameter section	Real data from VPS
15	`/intel`	Check feed cards	Shows ONLINE/OFFLINE from API
16	`/intel`	Enter 8.8.8.8, LOOKUP	Shows reputation results
17	`/intel`	Click SYNC FEEDS	Spinner + IOC count updates
18	`/network`	Check flow table	Live flows with ML Score
19	`/network`	Check TrafficTimeline	Non-empty chart
20	`/forensics`	Click upload area	File picker opens
21	`/forensics`	Upload a .pcap file	Upload succeeds, progress shown
22	`/reports`	Click Generate Report	PDF generated, appears in list
23	`/reports`	Click Download PDF	PDF downloads
24	`/admin`	Check Audit Log card	Shows real count from VPS
25	`/admin`	Check LLM Budget card	Shows real spend from VPS
26	`/hunt`	Type search query	Real flows returned
27	`/hunt`	Click Export CSV	CSV file downloads
28	All protected pages	Remove token, reload	Redirects to /login

7. Architecture Compliance

All changes must comply with:

1. **No Tailwind CSS** — Vanilla CSS + CSS Variables only ☑
2. **JetBrains Mono for data** — `var(--font-data)` for all metrics, IPs, scores ☑
3. **Inter for UI** — `var(--font-ui)` for labels, descriptions ☑
4. **Outfit for headings** — `var(--font-heading)` for page titles ☑

5. **GlassPanel containers** — All card/panel containers use GlassPanel ☑
6. **Dark theme** — Deep Space palette with glassmorphism ☑
7. **Design system tokens** — All colors use CSS variables ☑
8. **No prohibited technologies** — No Kafka, K8s, Elasticsearch, MongoDB ☑
9. **Ensemble weights locked** — Frontend displays, never modifies ☑
10. **10 modules only** — No new routes/modules added ☑

Day 18 Plan — Frontend Overhaul & VPS Integration To be appended to worklog after completion